HW3: FARKLE

This is the continuation of my Farkle Project made in Java from HW 1 and 2! It contains all of the basic functionality and now some quality-of-life improvements to the game. It also includes a brand new multiplayer hotseat with spearate scoring and Farkle endgame capabilities.

By Jett Marleau

PROCESS

Class and design Structure

This branch of the project relies heavily on the preexisting architecture from HW 1 and 2, but now adds much clearer menu printing, and a new Player class that facilitates a multiplayer experience and helps create a functioning scoreboard.

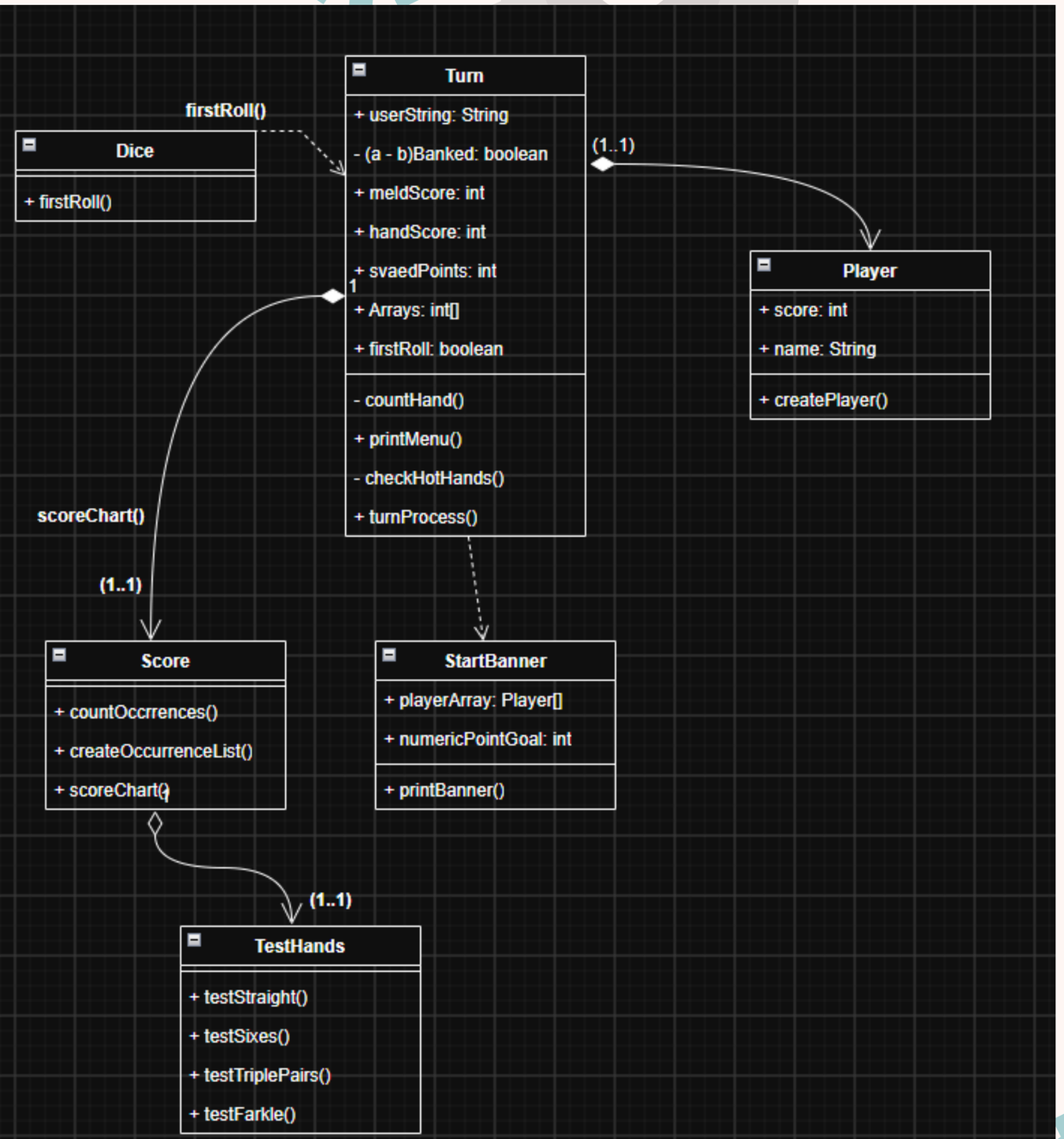
Unit Testing

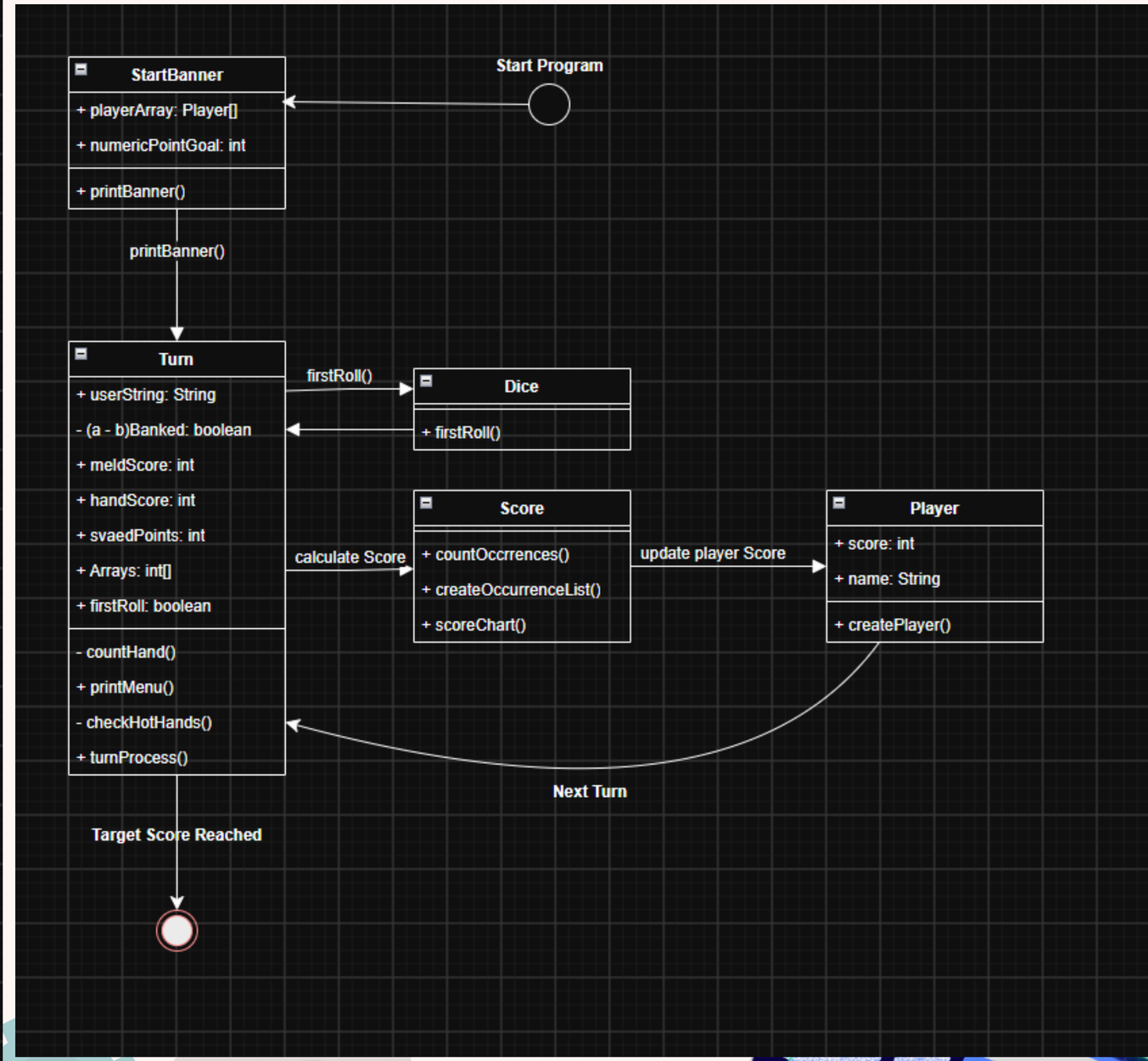
Unit Testing is the same as HW 2, test arrays with different dice combos are seeded into the game for each player, and they must all validate that scoring is correctly processed.

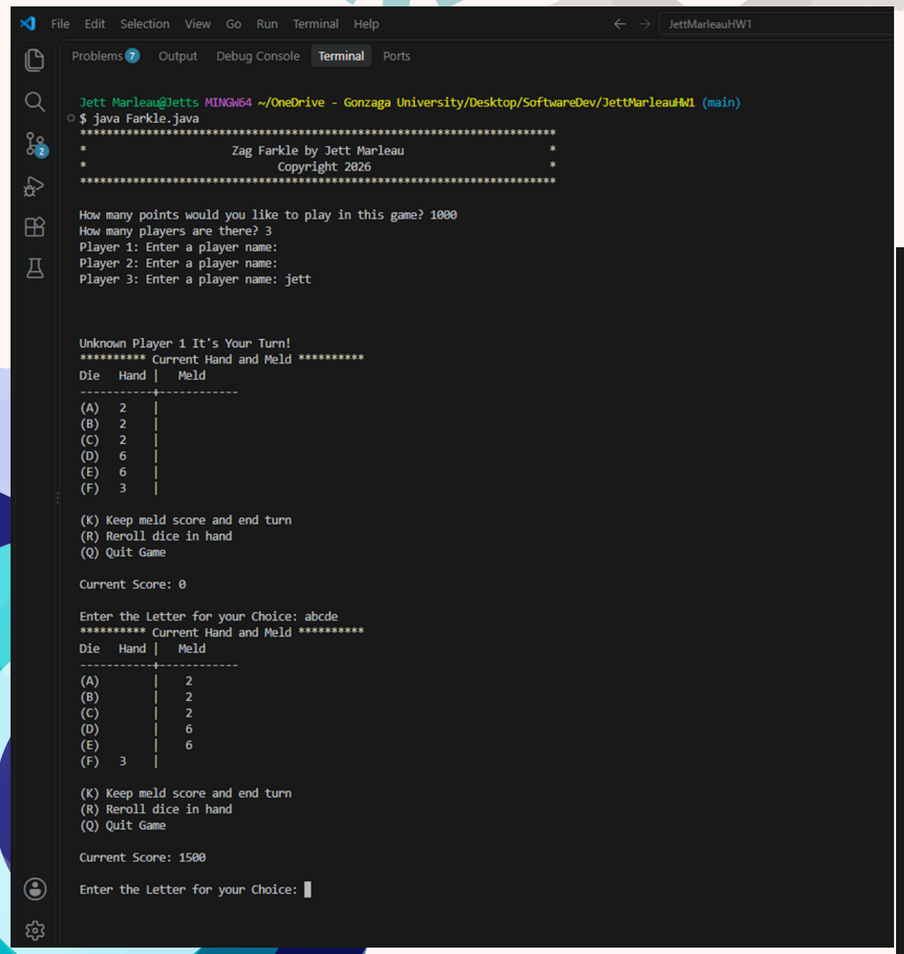
Challenges and Future Changes

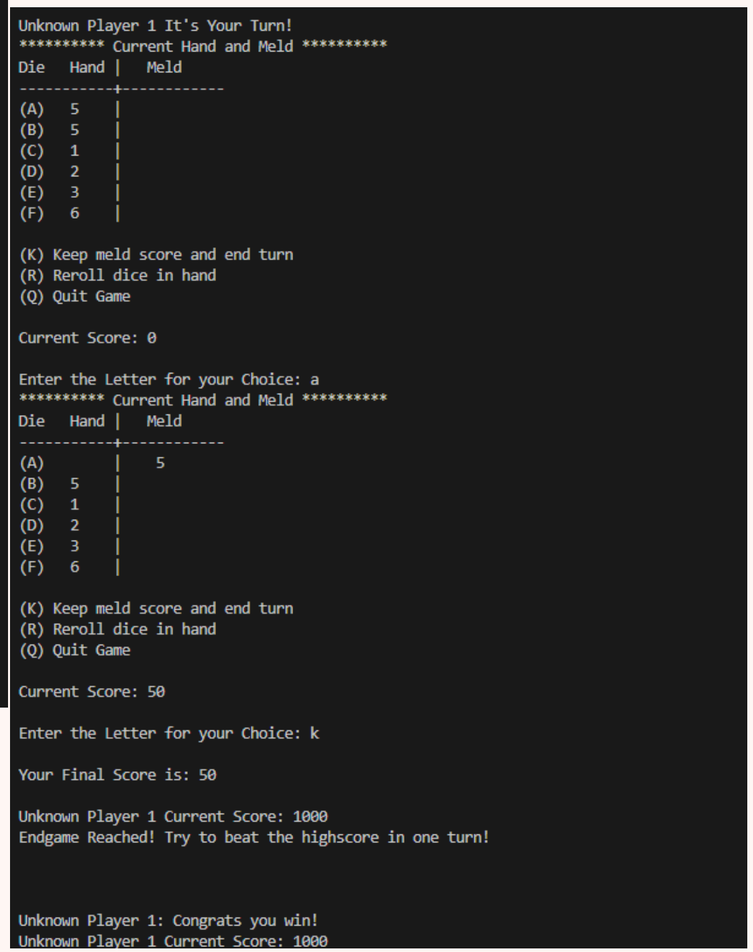
Finding bugs within the multiplayer system was difficult as their were multiple scoring paths and much mored nuanced behavoir than in single player. This also slowed down user testing as each player's turn needs to be taken. With more time I would love to tidy up the menu even more for more clarity and readablity.




Turn
+ userString: String
- (a - b)Banked: boolean
+ meldScore: int
+ handScore: int
+ svaedPoints: int
+ Arrays: int[]
+ firstRoll: boolean
- countHand()
+ printMenu()
- checkHotHands()
+ turnProcess()
firstRoll()
Dice
+ firstRoll()
(1..1)
Player
+ score: int
+ name: String
+ createPlayer()
1
scoreChart()
(1..1)
Score
+ countOccrrences()
+ createOccurrenceList()
+ scoreChart()
StartBanner
+ playerArray: Player[]
+ numericPointGoal: int
+ printBanner()
(1..1)
TestHands
+ testStraight()
+ testSixes()
+ testTriplePairs()
+ testFarkle()


StartBanner
+ playerArray: Player[]
+ numericPointGoal: int
+ printBanner()
Start Program
printBanner()
Turn
+ userString: String
- (a - b)Banked: boolean
+ meldScore: int
+ handScore: int
+ svaedPoints: int
+ Arrays: int[]
+ firstRoll: boolean
- countHand()
+ printMenu()
- checkHotHands()
+ turnProcess()
firstRoll()
Dice
+ firstRoll()
calculate Score
Score
+ countOccrrences()
+ createOccurrenceList()
+ scoreChart()
update player Score
Player
+ score: int
+ name: String
+ createPlayer()
Next Turn
Target Score Reached


File Edit Selection View Go Run Terminal Help
JettMarleauHW1
Problems 7 Output Debug Console Terminal Ports
Jett Marleau@Jetts MINGW64 ~/OneDrive - Gonzaga University/Desktop/SoftwareDev/JettMarleauHW1 (main)
$ java Farkle.java
****
* Zag Farkle by Jett Marleau *
* Copyright 2026 *
****
How many points would you like to play in this game? 1000
How many players are there? 3
Player 1: Enter a player name:
Player 2: Enter a player name:
Player 3: Enter a player name: jett
Unknown Player 1 It's Your Turn!
********** Current Hand and Meld **********
Die Hand | Meld
(A) 2 |
(B) 2 |
(C) 2 |
(D) 6 |
(E) 6 |
(F) 3 |
(K) Keep meld score and end turn
(R) Reroll dice in hand
(Q) Quit Game
Current Score: 0
Enter the Letter for your Choice: abcde
********** Current Hand and Meld **********
Die Hand | Meld
(A) | 2
(B) | 2
(C) | 2
(D) | 6
(E) | 6
(F) 3 |
(K) Keep meld score and end turn
(R) Reroll dice in hand
(Q) Quit Game
Current Score: 1500
Enter the Letter for your Choice:


Unknown Player 1 It's Your Turn!
********** Current Hand and Meld **********
Die Hand | Meld
(A) 5 |
(B) 5 |
(C) 1 |
(D) 2 |
(E) 3 |
(F) 6 |
(K) Keep meld score and end turn
(R) Reroll dice in hand
(Q) Quit Game
Current Score: 0
Enter the Letter for your Choice: a
********** Current Hand and Meld **********
Die Hand | Meld
(A) | 5
(B) 5 |
(C) 1 |
(D) 2 |
(E) 3 |
(F) 6 |
(K) Keep meld score and end turn
(R) Reroll dice in hand
(Q) Quit Game
Current Score: 50
Enter the Letter for your Choice: k
Your Final Score is: 50
Unknown Player 1 Current Score: 1000
Endgame Reached! Try to beat the highscore in one turn!
Unknown Player 1: Congrats you win!
Unknown Player 1 Current Score: 1000


Jett Marleau@Jetts MINGW64 ~/OneDrive - Gonzaga University/Desktop/SoftwareDev/JettMarleauHW1 (main)
$ java Farkle.java
Die Hand | Meld
(A) 2 |
(B) 1 |
(C) 4 |
(D) 5 |
(E) 5 |
(F) 4 |
(K) Keep meld score and end turn
(R) Reroll dice in hand
(Q) Quit Game
Current Score: 0
Enter the Letter for your Choice: b
********** Current Hand and Meld **********
Die Hand | Meld
(A) 2 |
(B) | 1
(C) 4 |
(D) 5 |
(E) 5 |
(F) 4 |
(K) Keep meld score and end turn
(R) Reroll dice in hand
(Q) Quit Game
Current Score: 100
Enter the Letter for your Choice: k
Your Final Score is: 100
jett Current Score: 400
bob Current Score: 200
Unknown Player 3 Current Score: 800
Unknown Player 4 Current Score: 100
Unknown Player 3: Congrats you win!
jett Current Score: 400
bob Current Score: 200
Unknown Player 3 Current Score: 800
Unknown Player 4 Current Score: 100
Jett Marleau@Jetts MINGW64 ~/OneDrive - Gonzaga University/Desktop/SoftwareDev/JettMarleauHW1 (main)
$